Predicting the long-term impact of AI assistants on collective intelligence

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Abstract

Artificial intelligence (AI) assistants can improve individual performance but may weaken unaided performance and limit interest diversity, raising questions about their long-term effects on collective intelligence. Here, we consider an evolutionary model in which individuals participate in a prediction task, consult AI, and imitate better-performing peers. Classic reward structures that promote collective intelligence without AI can fail under "Chatbot AI", whose predictions depend on users' interests, either by reducing interest diversity or propagating bias. Directly rewarding or penalizing AI use does not reliably mitigate these effects. Instead, a balanced incentive that preserves minority expertise and rewards collective contributions can promote collective intelligence, but an additional incentive for rewarding unaided performance is needed to restore unaided collective accuracy. Our framework offers a quantitative explanation for emerging evidence of AI's societal impact and clarifies its long-term implications.

Introduction

Collective intelligence (CI) describes a phenomenon in which a group's ability to solve problems, make decisions, or adapt to changing environments exceeds each member's ability Galton (1907); Lorge et al. (1958); Yaniv and Milyavsky (2007); Woolley et al. (2010); Hommes et al. (2005). This phenomenon, also referred to as wisdom of crowds, has thus served as a central mechanism underlying the success of diverse organizations, ranging from animal groups Dreyer et al. (2025); Simons (2004); Berdahl et al. (2013) and human societies Woolley et al. (2010); Riedl et al. (2021) to machine-learning systems Breiman (1996, 2001). Understanding the conditions that promote and maintain CI therefore remains a fundamental question across social sciences Mann and Helbing (2017), evolutionary biology Wang et al. (2025), and computer science Werfel (2026). Addressing this question has become particularly timely as artificial intelligence (AI) has begun to reshape how individuals acquire information, make decisions, and contribute to collective outcomes Cui and Yasseri (2024).

The recent emergence of generative AI and large language models (LLMs) has changed the way individuals engage in various cognitive tasks, ranging from everyday tasks Noy and Zhang (2023); Brynjolfsson et al. (2025) to scientific discovery Lu et al. (2026) and creative writing Doshi and Hauser (2024). In particular, while the use of generative AI models, such as ChatGPT, has been rapidly increasing Bick et al. (2026); Sajadieh et al. (2026), the potential long-term impact of AI assistants on human society, especially CI, remains uncertain. Some argue that AI can promote CI by enhancing individual expertise Vaccaro et al. (2024) and opinion aggregation Burton et al. (2024); Tessler et al. (2024). Conversely, empirical evidence indicates that the use of AI can limit the diversity of opinions Fügener et al. (2021); Doshi and Hauser (2024); Lee and Chung (2024); Meincke et al. (2025), a key requirement for the emergence of CI Hong and Page (2004); Lorenz et al. (2011); Aplin et al. (2014); Becker et al. (2017). We expect these AI-driven changes to shape long-term social evolution because successful individuals can influence which interests, beliefs, and patterns of AI use become prevalent in a population.

However, our ability to predict the long-term impact of AI assistants on human CI is hampered by the apparent discrepancy in the incentives that shape AI use. On one hand, many industrial sectors are pushing towards increased AI use, where the number of "tokens" used is often perceived as a proxy for productivity Bousquette (2026); Keary (2026). On the other hand, many academic institutions appear to discourage AI use, reflecting concerns about plagiarism, misinformation, and limited independence Thorp (2023); Chawla (2026). Such discrepancy raises a question about whether directly incentivizing or penalizing AI use will benefit or harm human CI. More broadly, previous work identified incentive structures that promote CI through social learning Wang et al. (2025), but it remains unclear whether these structures remain effective in the presence of AI assistants. Thus, a key challenge is to identify robust incentive structures that retain the benefits of AI while limiting risks such as

the loss of diversity in individual opinions, thereby allowing for the maintenance of CI under AI use.

To address these questions, we extend previous approaches for modeling CI, in which each individual participates in a prediction task and imitates better-performing peers Hong et al. (2012); Mann and Helbing (2017); Wang et al. (2025). Previous work modeled the prediction target Y as a linear combination of underlying causal factors, $X_1, X_2, \ldots, X_M$, (Fig. 1A) and assumed that each individual makes an independent prediction from a single observed factor of their interest e_A and their belief c_A: $Y_A = c_A X_{e_A}$ (Fig. 1B). Here, we further allow each individual to independently consult with AI assistants, which can also make their own predictions Y_A^{AI}, and revise their predictions by taking the weighted average according to their reliance on AI β_A: $Y_A^{\mathrm{re}} = \beta_A Y_A^{\mathrm{AI}} + (1-\beta_A) Y_A$ (Fig. 1B). In doing so, we consider two forms of AI: Omniscient AI constructs predictions using all causal factors, whereas Chatbot AI responds only about the factor that the user is interested in (Fig. 1B). These AI predictions can include both systematic biases and stochastic errors (Fig. 1B). Then, these revised predictions are aggregated to form a collective prediction $\hat{Y}^{\mathrm{re}}$, which is then used to evaluate the collective accuracy of the population Wang et al. (2025): $1-\mathbb{E}[(Y-\hat{Y}^{\mathrm{re}})^2]/\mathbb{E}[(Y-\alpha_0)^2]$. For Omniscient AI, we simply average revised predictions (namely, "averaging"). For Chatbot AI, we average within interests and sum across represented interests (namely, "clustering"). Hereafter, we say that CI emerges when the collective accuracy reaches a near-perfect level.

Individuals receive payoffs π_A based on their performance and imitate the interests, beliefs, and AI reliance of peers with higher expected payoffs, allowing the population to evolve over time (Fig. 1C). Following previous work Wang et al. (2025), we consider two incentive structures that can promote CI in the absence of AI when paired with an appropriate aggregation rule. First, the feedback structure gives higher payoffs to individuals whose predictions are aligned with the collective prediction error, rather than directly rewarding individual predictive accuracy: $\pi_A = Y_A^{\mathrm{re}}(Y - \hat{Y}^{\mathrm{re}})$. Second, the niche-expert structure gives higher payoff to individuals who have both better predictive accuracy and a minority interest: $\pi_A = -\rho_{e_A}(Y_A^{\mathrm{re}} - Y)^2$, where ρ_{e_A} represents the proportion of individuals sharing the same interest. We assume that prediction events occur on a substantially faster timescale than social learning, which allows us to calculate the expected payoff $\mathbb{E}[\pi_A]$ of each player analytically and simulate social learning. Full details of the model and derivations of the collective prediction, expected payoffs, and collective accuracy under the two AI models are provided in Materials and Methods as well as Supplementary Text.

We begin by characterizing the dynamics of CI with AI under two classic incentive structures from prior work Wang et al. (2025). We show that the outcome depends on the information available to AI: Omniscient AI can preserve or improve collective accuracy, whereas Chatbot AI can undermine CI by limiting interest diversity or propagating systematic AI bias. Directly incentivizing or penalizing AI use do not reliably correct these effects and can further reduce collective accuracy.

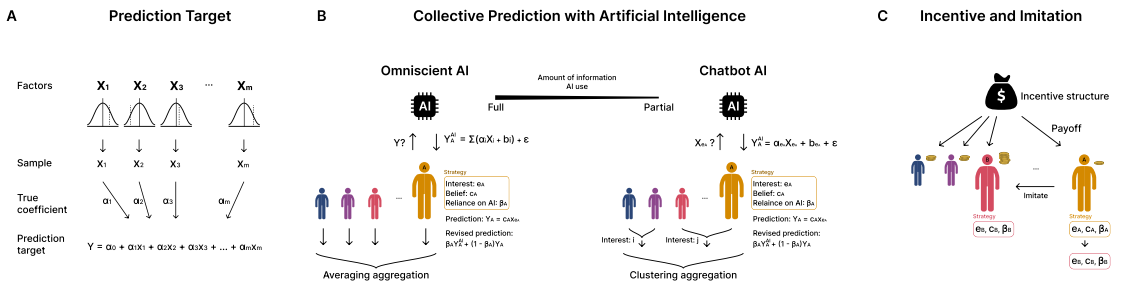


Figure 1: **A schematic diagram of the model for individual and collective prediction in the presence of AI assistants.** (A) The prediction target is modeled as a linear combination of causal factors, $X_1, X_2, \ldots, X_M$: $Y = \alpha_0 + \sum_{i=1}^{M} \alpha_i X_i$. (B) Each player is characterized by three strategies: interest e_A, belief c_A, and reliance on AI β_A, where where $e_A \in \{0, \ldots, M\}$ and $X_0 = 1$. Players make predictions based on their interests and beliefs and subsequently revise them with AI assistance. We consider two types of AI, Chatbot AI and Omniscient AI, which differ in the amount of information available for generating predictions. Chatbot AI makes predictions based on user's interest: $Y_A^{\text{AI}} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon_A$. Omniscient AI makes predictions based on all causal factors: $Y_A^{\text{AI}} = \sum_{i=0}^{M} (\alpha_i X_i + b_i) + \epsilon_A$, where $X_0 = 1$. For both cases, we assume $\epsilon_A \sim \mathcal{N}(0, \tau)$. (C) Players receive payoffs according to the incentive structure, and the population evolves through imitation of peers with higher payoffs.

Finally, we leverage these findings to propose a balanced incentive structure that preserves minority expertise while rewarding contributions to collective accuracy, thereby restoring CI under Chatbot AI. However, even when AI-assisted CI is restored, the population's unaided collective accuracy remains suboptimal, requiring additional incentives that explicitly reward unaided predictive ability.

Classic incentive structures for collective intelligence can fail under Chatbot AI

First, consistent with Wang et al. (2025), previously proposed incentive structures allow collective intelligence to emerge in the absence of AI (Fig. 2A,C,D). One exception is the use of simple averaging with niche-expert incentive structures, under which the collective accuracy of the population remains near zero. This is because each individual accurately predicts the contribution of their factor of interest, but averaging these partial predictions dilutes the factor-specific contributions instead of reconstructing the full prediction target (Fig. S1).

The use of Omniscient AI can either preserve CI under the feedback incentive structure (Fig. 2A) or promote higher collective accuracy under the niche-expert incentive structure (Fig. 2B). These results are robust across different levels of AI

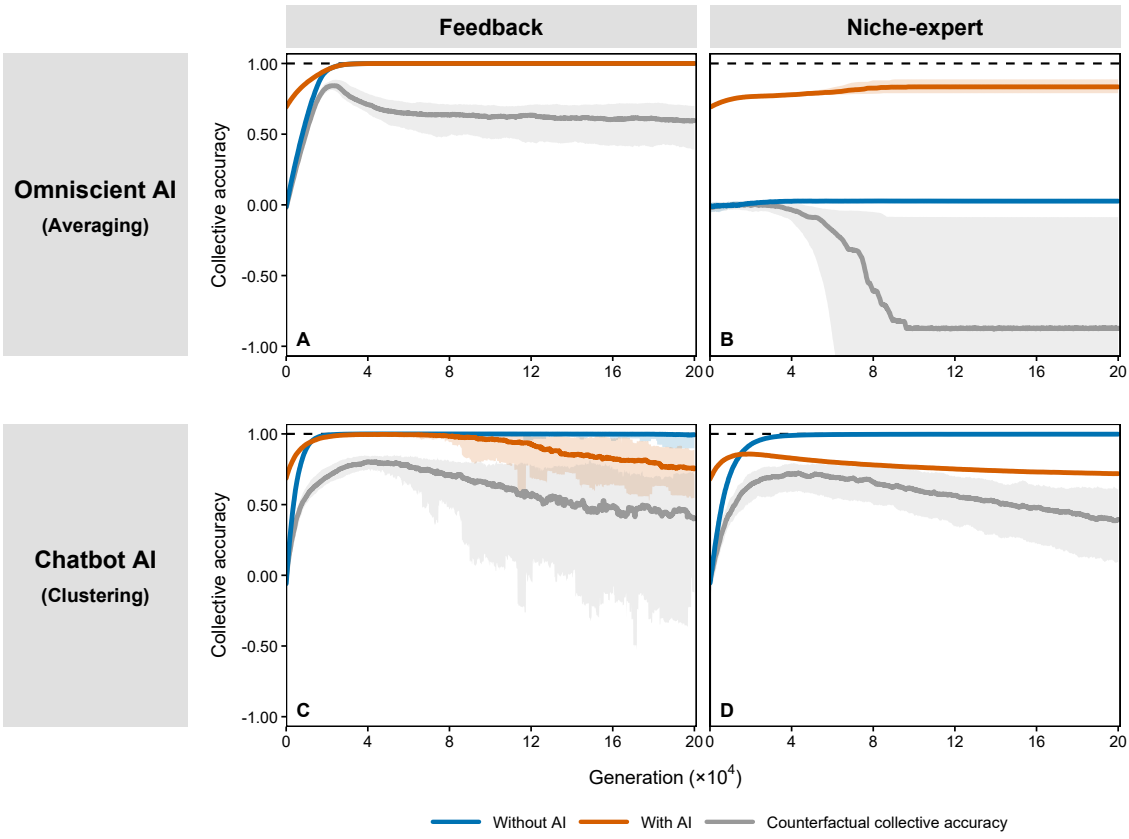


Figure 2: **Comparison of collective accuracy with and without AI.** Trajectories of collective accuracy when the group is assisted by the Omniscient AI (A,B) and Chatbot AI (C,D). Here, feedback (A,C) and niche-expert (B,D) incentive structures are considered. Blue lines represent trajectories of collective accuracy without AI assistance. Orange lines represent trajectories of collective accuracy with AI assistance. Gray lines represent trajectories of counterfactual collective accuracy, computed from players' unrevised predictions, indicating the group's predictive accuracy if AI assistance were unavailable at the given generation. Each condition was simulated for 30 independent replicates with randomly initialized player populations. Lines show the median across replicates, and shaded areas indicate the 95% simulation interval. Intervals extending below -1 are truncated by the y-axis limits for visual clarity. Prediction target parameters: $M = 50$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Evolutionary parameters: $s = 50$ and $\mu = 0$. Population parameters: $N = 10{,}000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$ and $\tau = 0.3$.

accuracy (Fig. S2) and can also be characterized analytically in the large-population limit (Supplementary Text).

Under the feedback incentive structure, the use of Chatbot AI leads to a loss of diversity in the population's interest, making it impossible to account for effect of some causal factors (Fig. S2, S3). This causes an increase in collective variance and a stochastic decrease in collective accuracy (Fig. 2C; Fig. S3). We note that the loss of interest diversity may arise even in the absence of AI in finite populations (Fig. S3), but the use of Chatbot AI makes the group more vulnerable to such loss (Fig. S4). Under the niche-expert incentive structure, the population becomes overly reliant on AI, causing the collective prediction to inherit the AI's bias (Fig. S3). This leads to an increase in collective bias and a decrease in collective accuracy (Fig. 2; Fig. S3). Interestingly, the niche-expert structure can instead reduce reliance on AI when the target intercept, α_0, and AI bias have opposite signs (Figs. S2,S5). In this case, the population reduces its reliance on AI assistance, allowing CI to emerge eventually. However, the emergence of CI can be substantially delayed compared with the population without AI (Fig. S5).

To investigate whether the group can retain its predictive ability when AI assistance suddenly becomes unavailable, we quantify counterfactual collective accuracy based on unrevised predictions, retaining the population's evolved interests and beliefs. Across all scenarios, counterfactual collective accuracy is lower than the accuracy of the corresponding AI-assisted population and the accuracy of populations that evolved without AI (Fig. 2). Thus, the accuracy achieved with AI assistance does not necessarily reflect the population's ability to make equally accurate predictions without it.

Direct incentives and penalties for AI do not reliably improve collective accuracy

Given the strong influence of AI on collective outcomes, a straightforward intervention is to directly incentivize or penalize the use of AI. To investigate whether such direct incentives or penalties can promote CI, we modify the two incentive structures by adding a payoff term proportional to individual reliance on AI:

$$\text{Feedback/Niche-expert} + \lambda\beta_A \tag{1}$$

where λ represents the strength of incentive ($\lambda > 0$) or penalty ($\lambda < 0$), and β_A represents individual reliance on AI (Supplementary Text).

We find that direct incentives and penalties on AI use do not consistently improve collective accuracy (Fig. 3). When Omniscient AI is used under the feedback structure, the population retains collective intelligence regardless of the intervention strength (Fig. 3A), despite large changes in reliance on AI (Fig. 3B). In other scenarios, directly incentivizing or penalizing AI use causes individuals to imitate others based on their AI use rather than their predictive accuracy (Fig. 3C,E,G). When AI

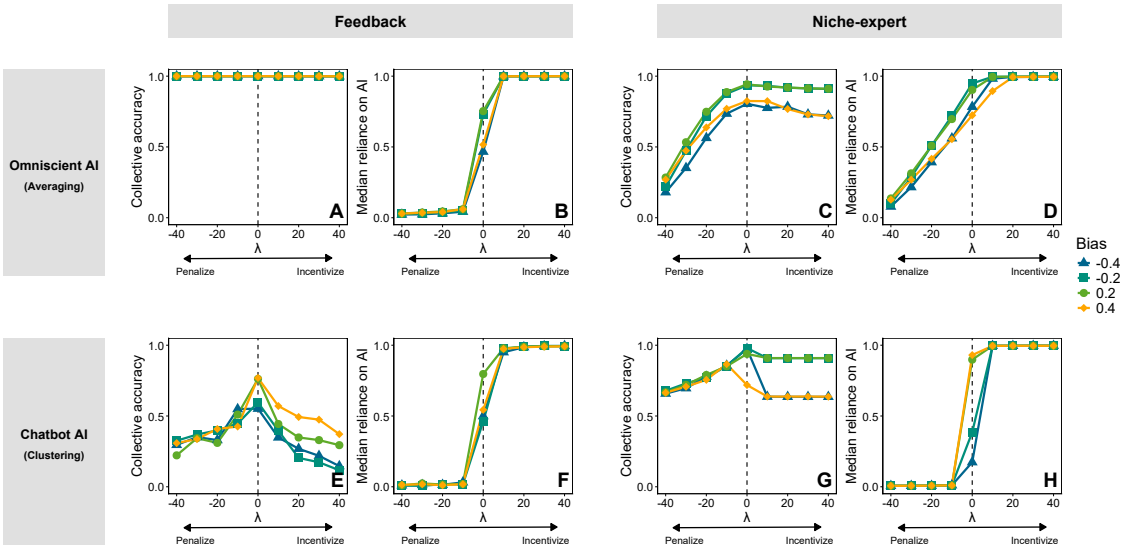


Figure 3: **Collective accuracy under the influence of direct intervention on AI use.** Collective accuracy and median reliance on AI under varying strengths of intervention (λ) across different AI types (Omniscient AI, A–D and Chatbot AI, E–H) and incentive structures (Feedback, A,B,E,F, and Niche expert, C,D,G,H). Positive λ represents an incentive for AI use, whereas negative λ represents a penalty for AI use. Four AI bias levels are shown: $b_i = \pm 0.2$, representing higher-accuracy AI, and $b_i = \pm 0.4$, representing lower-accuracy AI. Each condition was simulated for 30 independent replicates with randomly initialized player populations. Collective accuracy (A,C,E,G) and median reliance on AI (B,D,F,H) are averaged over generations 190,001–200,000 within each replicate, and points represent the median across replicates. Prediction target parameters: $M = 50$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Evolutionary parameters: $s = 50$ and $\mu = 0$. Population parameters: $N = 10{,}000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A–D), $c_A \sim \mathcal{N}(0, 5^2)$ for (E–H), and $\beta_A \sim U(0, 1)$. AI parameter: $\tau = 0.3$.

bias is relatively large, a weak penalty can lead to higher collective accuracy, but this improvement is restricted to a narrow parameter regime (Fig. 3G). Overall, these results indicate that directly rewarding or penalizing AI use may exacerbate rather than mitigate its effects on collective accuracy.

Balanced incentives promote collective intelligence with AI

The contrasting failures of the feedback and niche-expert structures suggest combining their complementary roles. Here, we propose a "balanced" incentive structure, modeled as the average of the feedback and niche expert incentives (Supplementary

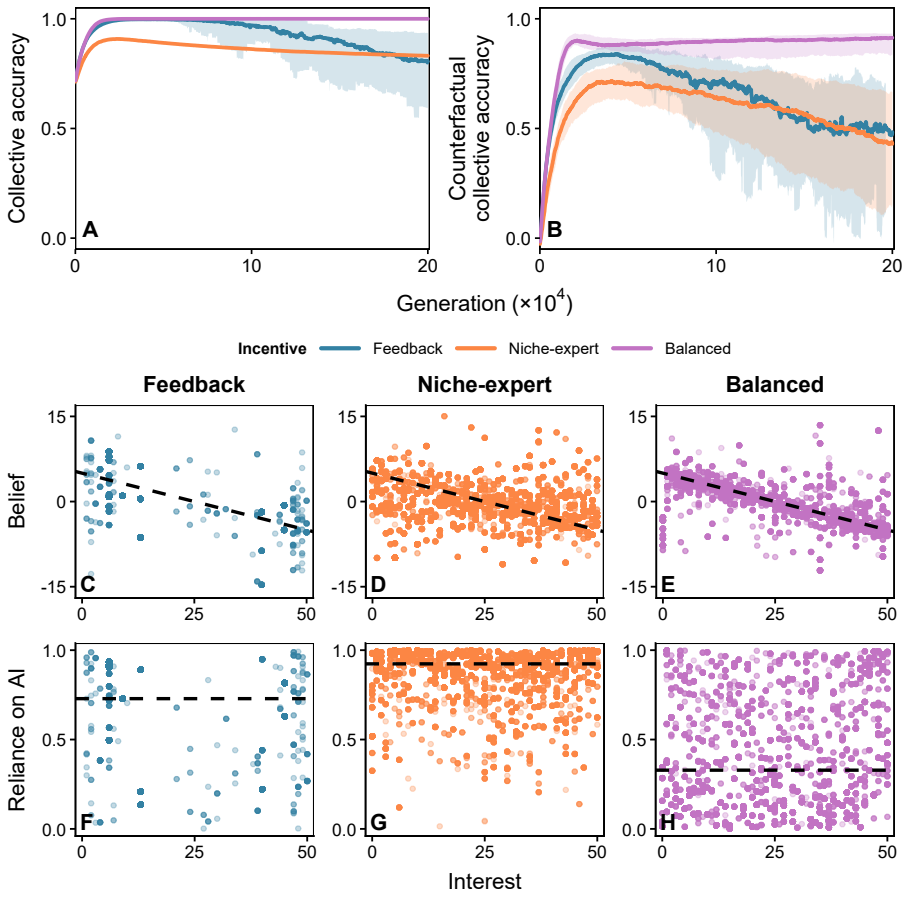


Figure 4: **Balanced structure can promote collective intelligence.** Trajectories of collective accuracy (A) and counterfactual collective accuracy (B) under three incentive structures (feedback, niche-expert, and balanced) with Chatbot AI. Each incentive structure was simulated for 30 independent replicates with randomly initialized player populations. Lines show the median across replicates, and shaded areas indicate the 95% simulation interval. Distributions of individual interest and belief (C-E) and of interest and reliance on AI (F-H) after 200,000 generations in one representative replicate. Each point represents an individual. Dashed lines in panels C-E represent the true coefficients (α_i) of the prediction target. Dashed horizontal lines in panels F-H indicate the median reliance on AI. Prediction target parameters: $M = 50$, $\alpha_i = 5 - 0.2i$, and $\sigma_i \sim U(0, 3)$. Evolutionary parameters: $s = 50$ and $\mu = 0$. Population parameters: $N = 10{,}000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$ and $\tau = 0.3$.

Text).

The core idea behind the balanced structure is to maintain interest diversity and prevent collective bias. On one hand, the feedback incentive structure is prone to

the loss of interest diversity but can prevent collective bias by rewarding predictions aligned with the collective prediction error. On the other hand, the niche-expert structure can maintain interest diversity because it directly incentivizes minority interests. However, it does not directly reward the correction of collective bias, because individuals sharing the same interest are rewarded according to their individual predictive performance.

Therefore, when the two incentive structures are combined, the population achieves CI by maintaining interest diversity while limiting collective bias (Fig. 4A; Fig. S6). Under the balanced structure, individuals remain more evenly distributed across causal factors, and their beliefs lie closer to the true coefficients than under either incentive structure alone (Fig. 4C–E). The balanced structure also produces higher counterfactual collective accuracy than the feedback or niche-expert structure (Fig. 4B; Fig. S6). These patterns are robust to balancing weights (Fig. S7), and are driven by the emergent property of the balanced structure, under which the population increases its reliance on AI as AI becomes more accurate (Fig S8).

However, counterfactual collective accuracy remains suboptimal even under the balanced structure because payoffs are based only on revised predictions. An analogous incentive based on unrevised predictions raises counterfactual collective accuracy to near-perfect levels but does not calibrate reliance on AI, causing collective accuracy to decline as the magnitude of AI bias increases. Combining the two structures avoids this trade-off and maintains near-perfect collective accuracy both with and without AI across the range of AI bias examined (Supplementary Text; Fig. S9).

Discussion

The ongoing race towards the development of "better" AI continues to pose questions about its societal benefits and risks Bender et al. (2021); Burton et al. (2024); Cui and Yasseri (2024). Here, we used mathematical models of evolutionary dynamics to investigate how AI use and incentive structures jointly affect the emergence of CI. We find that an "Omniscient" AI model, which incorporates information about all relevant causal factors, can improve collective accuracy, whereas a "Chatbot" AI model, whose outcome depends on the user's interest, can impair it. These contrasting outcomes arise because AI assistance affects not only individual predictions, but also the population's interest diversity, beliefs, and reliance on AI. We also show that directly incentivizing or penalizing AI use, independent of individual's performances, can further limit the collective accuracy of the population. Instead, a balanced incentive structure can maintain CI by promoting diversity and limiting bias.

Our analysis reveals that AI can prevent the emergence of CI through two different mechanisms. First, AI use can cause individuals to concentrate on a subset of the factors that determine the prediction target, thereby reducing diversity in their interests. This result is broadly consistent with recent empirical studies showing that AI assistance can improve individual performance but limit the diversity of outputs,

in domains ranging from scientific research Hao et al. (2026) to creative writing Doshi and Hauser (2024). Second, over-reliance on AI can cause the population to inherit inherent biases in AI's predictions Fügener et al. (2021); Vicente and Matute (2023), causing the prediction accuracy to decrease eventually. This mechanism can prevent the emergence of CI even when the interest diversity is maintained because biases in AI predictions can be amplified when individual predictions are aggregated. Thus, preserving the diversity of ideas will not be sufficient to promote CI. Instead, AI use need to be calibrated to prevent systematic AI biases from becoming collective biases Buçinca et al. (2021).

Our analysis of counterfactual collective accuracy also raises an important concern for future AI use: while AI use may improve the apparent performance of individuals and, in some cases, the population as a whole, it can simultaneously reduce the population's ability to make accurate predictions without AI. These results echo findings from a recent field experiment, which demonstrated that students, who seemingly performed better in solving mathematics problems under AI assistance, performed worse on their own than those who did not have access Bastani et al. (2025). Our results, however, also suggest that this loss of unaided performance is not inevitable. Incorporating incentives based on individuals' predictions without AI assistance can improve counterfactual collective accuracy without losing overall collective accuracy. These findings highlight the need to evaluate AI tools not only by the performance gains they provide during use, but also by whether individuals and populations retain the ability to perform without them Melumad and Yun (2025); Ke et al. (2026). Making this distinction clear will be particularly important for education, where AI may help students complete a task without helping them learn how to do it on their own Bastani et al. (2025); Melumad and Yun (2025).

As generative AI and large language models are already generating substantial benefits, many organizations, both industrial and academic, are increasingly seeking to improve their collective performance by regulating AI use Thorp (2023); Bousquette (2026); Chawla (2026); Keary (2026). Such regulations may take direct forms, including AI-use policies or token-based leaderboards, or may operate indirectly through other incentives, such as productivity rewards or the social stigma associated with AI use Reif et al. (2025). Our model suggests that incentivizing or penalizing AI use is unlikely to induce appropriately calibrated reliance on AI or to promote CI. Incentivizing AI use could lead to overreliance, causing groups to propagate AI biases, whereas penalizing AI use could lead to underreliance, preventing groups from using AI even when it performs well in the relevant domain. Our results instead show that CI and appropriate reliance on AI can be promoted by rewarding diversity and contributions to collective accuracy Hong and Page (2004); Lorenz et al. (2011); Mann and Helbing (2017), rather than by directly rewarding or penalizing AI use.

There are several limitations to our analyses. First, our model allows for social learning among humans, but AI's inherent ability remains fixed throughout the evolutionary process. In reality, AI can also learn from human-generated data Ouyang

et al. (2022). Such bidirectional learning could increase the similarity between human and AI errors and thereby reduce group performance Glickman and Sharot (2025). Second, we model AI errors using fixed, factor-specific additive biases and independent random errors. This assumption allows random errors to be reduced through aggregation, while systematic biases can still persist in the collective prediction. In reality, AI errors may have more complex structures, including biased estimates of factor effects and correlated errors across individuals, factors, or repeated interactions. Further empirical and theoretical work is therefore needed to understand how these error structures affect the emergence of CI. Third, in our main analyses, each AI model is paired with the aggregation rule that best matches its prediction structure, making it difficult to fully separate the effects of the AI model from those of aggregation. However, analyses of alternative pairings suggest that our primary aggregation choices are consistent with the distinct prediction structures of the two AI models, although the effects of AI assistance may also depend on how individual predictions are aggregated (Supplementary Text; Fig. S10). Finally, following established models of CI, we assume that the prediction target is a linear combination of independent causal factors. This assumption does not hold for many real-world tasks, where causal factors may be correlated or interact nonlinearly. Future studies should therefore investigate how nonlinear relationships between causal factors and prediction targets affect the interaction between AI assistance and the emergence of CI.

The rapid adoption of AI assistants is changing how individuals learn, work, and make decisions Tessler et al. (2024); Glickman and Sharot (2025); Sajadieh et al. (2026); Bick et al. (2026); Ke et al. (2026). While the immediate benefits and risks of these systems have been evaluated at the individual level Noy and Zhang (2023); Bastani et al. (2025), our study raises a broader concern that their widespread AI use may reshape collective intelligence of human society. Our results further underline the importance of designing reward structures that preserve minority expertise and prevent AI biases from persisting across the population Mann and Helbing (2017); Vicente and Matute (2023); Glickman and Sharot (2025); Wang et al. (2025). A detailed empirical understanding of how AI assistance reshapes diversity, expertise, and reliance at the population level will therefore be critical for predicting the dynamics of future of human collective intelligence.

Materials and Methods

Prediction target

Following previous models of CI, we model the prediction target as a linear combination of M independent causal factors Hong et al. (2012); Mann and Helbing (2017); Wang et al. (2025):

$$Y = \alpha_0 + \alpha_1 X_1 + \cdots + \alpha_M X_M, \tag{2}$$

where the coefficient α_i determines the magnitude and direction of the effect of factor i, and α_0 represents a global intercept. Each causal factor X_i is modeled as an independent Gaussian random variable with mean zero and standard deviation σ_i.

Players

We consider a well-mixed population $\mathcal{N}$ consisting of N players engaged in a prediction task Wang et al. (2025). In the absence of AI assistance, the strategy of player A is characterized by two variables: interest e_A and belief c_A. Player A observes the causal factor X_{e_A} and makes a prediction by multiplying the observed factor value by their belief:

$$Y_A = c_A X_{e_A}, \tag{3}$$

where e_A can take an integer value from 0 to M. When $e_A = 0$, the player does not observe any causal factor and instead makes a constant prediction based solely on their belief. We refer to this as a stubborn strategy because the player's prediction does not respond to the realized values of the causal factors. For notational convenience, we assume that $X_0 = 1$.

When the group is assisted by AI, each player's strategy additionally includes reliance on AI, denoted by β_A. We assume that each player consults the AI independently and revises their prediction by taking a weighted average of their own prediction and the AI-generated prediction:

$$Y_A^{\text{re}} = \beta_A Y_A^{\text{AI}} + (1 - \beta_A) Y_A, \tag{4}$$

where β_A ranges from 0 to 1. A value of $\beta_A = 1$ indicates complete reliance on AI, whereas $\beta_A = 0$ indicates that the player does not use the AI prediction.

AI models

We assume that the AI model has been trained on similar tasks and is therefore capable of making predictions based on causal factors. However, because the model is imperfect, its outputs are subject to systematic additive bias and stochastic error. The additive bias associated with factor i is denoted by b_i, and the stochastic error is denoted by ϵ. The AI is neither trained nor updated during the simulation, and its biases therefore remain fixed throughout the simulation. For simplicity, we assume $b_i = b$ for the main analysis for all i. The model formulation nevertheless allows biases to differ among factors.

We assume that the stochastic error follows a Gaussian distribution with mean zero and standard deviation τ. Each time a player consults the AI, an independent error is generated. Errors are therefore independent across players and across repeated consultations by the same player. Consequently, stochastic AI errors can be attenuated when independently generated predictions are aggregated across players. We therefore hold τ constant and vary the magnitude of the additive bias to manipulate the level of AI accuracy during our analyses.

We consider two extreme types of AI that differ in the amount of information available for prediction. The first is termed the Omniscient AI, which can observe all M causal factors and incorporate them into its prediction regardless of the user's interest. The prediction generated by the Omniscient AI for player A is expressed as follows:

$$Y_A^{\mathrm{AI}} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon. \tag{5}$$

The second is termed the Chatbot AI, which can use only the single causal factor X_{e_A} associated with the player's interest. The prediction generated by the Chatbot AI for player A is expressed as follows:

$$Y_A^{\mathrm{AI}} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon. \tag{6}$$

Aggregation rules

The revised individual predictions are aggregated to form a collective prediction. We use two aggregation rules proposed in previous work Wang et al. (2025), depending on the type of AI used by the group. First, when the group is assisted by the Omniscient AI, the AI component of each revised prediction contains information about all M factors, even though the individual's own prediction is based on only one factor. Each revised prediction can therefore be treated as an estimate of the entire target. Accordingly, we calculate the collective prediction by taking the simple average of the revised individual predictions:

$$\hat{Y}^{\mathrm{re}} = \frac{Y_1^{\mathrm{re}} + Y_2^{\mathrm{re}} + \cdots + Y_N^{\mathrm{re}}}{N}. \tag{7}$$

Second, when the group is assisted by the Chatbot AI, both the player's initial prediction and the AI's prediction are based on the same causal factor. Each revised prediction therefore contains information about only one factor and can be treated as an estimate of that factor's contribution to the target. We consequently use clustering aggregation.

Clustering aggregation is performed by grouping individuals according to their interests and calculating the mean prediction within each group. The collective prediction is then obtained by summing the mean predictions of all nonempty groups:

$$\hat{Y}^{\mathrm{re}} = \sum_{i:n_i>0} \frac{1}{n_i} \sum_{A\in\mathcal{A}_i} Y_A^{\mathrm{re}}, \tag{8}$$

where $\mathcal{A}_i$ represents the set of players interested in factor i, and $n_i = |\mathcal{A}_i|$ represents the number of players in that set.

When the group is not assisted by AI, collective predictions are computed from the unrevised individual predictions throughout the simulation. Both simple averaging and clustering can be used in the absence of AI. For fair comparison, we use the same aggregation rule as in the corresponding AI-assisted condition.

Incentive structures

Players receive payoffs based on the underlying incentive structure. We consider seven incentive structures: feedback structure, niche-expert structure, direct intervention on AI use of these two structures, and three different balanced structures.

The feedback structure does not directly reward individuals according to their own prediction accuracy. Instead, it rewards predictions that are aligned with the residual error of the collective prediction and therefore have the potential to move the collective prediction toward the true value. Given player A's revised prediction Y_A^{re} and the collective prediction $\hat{Y}^{\mathrm{re}}$, the payoff is expressed as

$$\pi_A = Y_A^{\mathrm{re}}(Y - \hat{Y}^{\mathrm{re}}). \tag{9}$$

The niche-expert structure assigns higher payoffs to individuals who make accurate predictions while having an underrepresented interest. Specifically, the squared prediction error of player A is weighted by $\rho_{e_A} = n_{e_A}/N$, the proportion of players interested in the same factor:

$$\pi_A = -\rho_{e_A}(Y_A^{\mathrm{re}} - Y)^2. \tag{10}$$

Previous research has shown that these two incentive structures can produce near perfect collective accuracy through social learning, except when the niche-expert incentive is combined with simple averaging Wang et al. (2025). They are therefore used as baseline structures for examining how AI assistance affects the emergence of CI.

To examine the effects of directly encouraging or discouraging AI use, we modify the feedback and niche-expert structures by adding a term proportional to the player's reliance on AI:

$$\pi_A = Y_A^{\mathrm{re}}(Y - \hat{Y}^{\mathrm{re}}) + \lambda\beta_A \tag{11}$$

$$\pi_A = -\rho_{e_A}(Y_A^{\mathrm{re}} - Y)^2 + \lambda\beta_A. \tag{12}$$

Here, λ represents the strength of the direct incentive or penalty applied to AI use. A positive value of λ rewards AI reliance, whereas a negative value penalizes it. When $\lambda = 0$, no direct incentive or penalty is applied, and the payoff functions reduce to the original feedback and niche-expert structures.

The balanced structure is defined as the weighted mean of the feedback and niche-expert payoff components, with balancing weight w:

$$\pi_A = (1 - w)Y_A^{\mathrm{re}}(Y - \hat{Y}^{\mathrm{re}}) - w\rho_{e_A}(Y_A^{\mathrm{re}} - Y)^2, \tag{13}$$

where $w \in [0, 1]$ determines the relative contribution of the feedback and niche-expert components. We set $w = 0.5$ for all analyses unless otherwise specified, such that the two components contribute equally to the balanced structure. The sensitivity of the results to w is examined in Supplementary Fig. S7.

To examine whether incentives based on unrevised predictions (i.e., predictions made without incorporating AI predictions) could overcome the limitations of the other incentive structures, we defined an unrevised balanced structure. The unrevised balanced structure is defined as the arithmetic mean of the feedback structure based on the unrevised collective prediction and the niche-expert structure based on unrevised individual predictions:

$$\pi_A = \frac{Y_A(Y - \hat{Y}) - \rho_{e_A}(Y_A - Y)^2}{2}, \tag{14}$$

where $\hat{Y}$ is the collective prediction based on unrevised individual predictions, and $Y_A(Y-\hat{Y})$ and $-\rho_{e_A}(Y_A-Y)^2$ represent the unrevised feedback and unrevised niche-expert components, respectively.

We further defined a combined balanced structure as the arithmetic mean of the balanced and unrevised balanced structures, with the balancing weight set to $w = 0.5$:

$$\pi_A = \frac{Y_A^{\mathrm{re}}(Y - \hat{Y}^{\mathrm{re}}) - \rho_{e_A}(Y_A^{\mathrm{re}} - Y)^2 + Y_A(Y - \hat{Y}) - \rho_{e_A}(Y_A - Y)^2}{4}. \tag{15}$$

Simulations

All simulations are conducted using an agent-based model with 50 causal factors ($M = 50$) and 10,000 players ($N = 10{,}000$). The parameters defining the prediction target are sampled at the beginning of the simulations and subsequently held fixed.

Coefficients α_i are sampled from a uniform distribution on $[-5, 5]$, except in the simulation presented in Figure 4, in which α_i is set to $5 - 0.2i$ for visualization. Standard deviations σ_i are sampled from a uniform distribution on $[0, 3]$.

Players' initial strategies are also sampled from predefined probability distributions. The distribution of individual belief c_A depends on the aggregation rule. Under simple averaging, each factor's contribution is scaled by the proportion of players interested in that factor. We therefore sample beliefs from a Gaussian distribution with mean zero and standard deviation 100, following the scaling used in previous work Wang et al. (2025). Under clustering aggregation, beliefs are sampled from a Gaussian distribution with mean zero and standard deviation 5. Because predictions are averaged separately within each interest group and then summed, no additional scaling of c_A is required. The initial interest of each player is sampled uniformly from the integers between 0 and 50, and reliance on AI is sampled from a uniform distribution on $[0, 1]$. To account for stochastic variation, we perform 30 independent simulation replicates with randomly initialized player populations for each condition presented in the main figures.

We assume that prediction events occur on a substantially faster timescale than social learning. In each generation, we therefore calculate the expected payoff $\mathbb{E}[\pi_A]$ of each player analytically rather than explicitly sampling prediction events. The expectation is taken over the distributions of the causal factors $X_1, \ldots, X_M$ and the

stochastic AI error ϵ, conditional on the current population strategies and the fixed target parameters. Closed form expression of expected payoffs are derived for all incentive structures and are presented in the Supplementary Text.

Each generation consists of an expected payoff calculation followed by one opportunity for pairwise social learning. Two distinct players, A and B, are randomly sampled, and player A imitated all three strategies of player B with probability

$$\frac{1}{1+\exp\left(s\left(\mathbb{E}[\pi_A]-\mathbb{E}[\pi_B]\right)\right)}, \tag{16}$$

where s represents the strength of social learning and was set to 50 in all simulations. We introduce a mutation rate μ, such that with probability μ, player A does not imitate player B but instead redrew all strategy components from the same distributions used to initialize the population. The mutation rate is set to zero in all simulations except those presented in Supplementary Fig. S4.

Unless otherwise specified, each simulation is run for 200,000 generations. When a result is represented by a single summary value rather than as a temporal trajectory, the metric was averaged over the final 10,000 generations.

To evaluate the dynamics of collective prediction, we define six summary metrics: collective accuracy, collective bias, collective variance, counterfactual collective accuracy, interest diversity, and median reliance on AI. At each population state, expectations and variances are evaluated analytically over the distributions of the causal factors and AI errors rather than estimated from sampled prediction events. Collective bias and collective variance constitute the bias–variance decomposition of the group level collective prediction error.

Collective accuracy is normalized relative to the baseline prediction α_0. A negative value indicates that the collective prediction performs worse than the baseline. Counterfactual collective accuracy is evaluated by setting $\beta_A = 0$ while retaining the evolved interests and beliefs of the AI-assisted population. In calculating interest diversity, we use the convention that $\rho_i \ln \rho_i = 0$ when $\rho_i = 0$ (Table **??**).

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